

DE AAR, South Africa

WMO: 685380

Lat: 30.6650S

Lon: 23.9928E

Elev: 4219

StdP: 12.59

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.0	33.5	3.8	7.8	60.5	7.7	9.5	60.7	31.4	60.8	26.0	55.8	7.8	150	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.1	95.0	60.1	92.9	60.1	90.8	59.8	67.8	79.9	66.5	78.6	65.4	77.9	14.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	107.4	70.7	63.3	102.0	70.2	61.8	96.6	69.7	34.8	79.8	33.7	77.9	32.7	78.0	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	23.8	21.1	DB	25.5	99.4	2.2	2.3	23.9	101.0	22.6	102.4	21.3	103.7	19.7	105.4	(4)
(5)			WB	23.9	69.2	2.1	1.7	22.3	70.5	21.1	71.4	19.9	72.4	18.3	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	75.4	75.0	70.9	63.0	56.6	50.2	49.8	53.6	59.9	65.5	68.9	73.5	(6)	
	DBStd	10.97	4.89	4.73	5.73	6.24	6.36	5.18	6.11	7.26	7.37	7.14	6.96	5.21	(7)	
	HDD50	222	0	0	0	2	20	61	81	49	9	0	0	0	(8)	
	HDD65	1985	2	2	18	108	266	444	472	358	184	84	42	5	(9)	
	CDD50	5131	787	699	649	391	223	66	74	159	306	482	567	728	(10)	
	CDD65	1422	325	281	202	47	4	0	0	4	31	101	159	268	(11)	
	CDH74	16772	3777	3075	2031	471	59	0	1	66	496	1330	2089	3377	(12)	
	CDH80	7112	1826	1405	772	84	2	0	0	5	124	462	859	1573	(13)	
(14) Wind	WSAvg	10.1	11.3	10.4	9.6	8.8	8.3	8.3	9.0	9.7	10.4	11.5	12.0	11.7	(14)	
(15) Precipitation	PrecAvg	13.0	1.4	2.0	2.2	1.5	0.7	0.6	0.4	0.6	0.5	1.0	1.2	0.9	(15)	
	PrecMax	26.2	7.5	9.4	6.5	4.7	2.3	3.4	2.9	3.3	3.4	7.0	4.5	4.7	(16)	
	PrecMin	6.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	4.7	1.5	2.0	1.5	1.1	0.7	0.7	0.7	0.9	0.8	1.3	1.0	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.9	96.6	93.6	87.3	78.9	70.7	71.9	79.7	88.6	92.0	94.0	97.0	(19)	
		MCWB	61.2	61.6	60.6	59.0	54.5	50.2	48.3	52.4	55.2	56.6	58.7	59.7	(20)	
	2%	DB	95.0	93.7	90.3	82.5	75.5	67.8	68.4	75.4	83.9	88.1	91.3	93.9	(21)	
		MCWB	60.6	61.6	60.2	58.3	53.3	48.7	48.0	50.4	54.1	56.6	58.1	59.2	(22)	
	5%	DB	92.4	91.3	87.6	79.6	72.7	65.4	66.2	72.4	79.8	85.2	88.7	91.3	(23)	
		MCWB	60.6	61.2	59.9	57.5	52.7	48.0	47.3	49.6	52.5	56.2	57.6	59.0	(24)	
	10%	DB	89.7	88.6	84.6	76.4	69.5	62.9	63.5	69.1	76.1	81.9	85.5	88.5	(25)	
		MCWB	60.7	61.1	59.5	56.3	52.1	46.9	46.5	48.1	51.1	55.2	57.0	58.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	69.3	68.0	64.3	58.8	53.5	52.4	55.8	59.5	62.9	65.7	67.3	(27)	
		MCDB	81.7	83.3	78.1	74.4	68.1	65.0	63.9	70.7	76.5	77.7	77.1	80.1	(28)	
	2%	WB	67.9	67.6	65.9	62.1	56.6	51.3	50.0	53.6	57.1	61.0	63.3	65.2	(29)	
		MCDB	79.9	80.3	77.2	73.7	67.8	62.0	63.8	68.4	74.2	77.0	76.4	78.6	(30)	
	5%	WB	66.4	66.5	64.5	60.5	54.9	49.7	48.5	51.4	55.2	59.2	61.5	63.8	(31)	
		MCDB	78.6	79.0	76.5	71.8	67.7	61.4	63.1	67.2	73.2	76.7	76.3	78.3	(32)	
	10%	WB	65.1	65.4	63.1	59.1	53.5	48.2	47.1	49.6	53.2	57.5	59.9	62.5	(33)	
		MCDB	78.3	77.9	75.4	70.2	65.8	60.2	61.5	65.8	71.5	75.8	77.0	77.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	26.1	24.8	23.8	21.9	21.2	21.8	22.9	24.4	26.1	26.7	27.0	27.2	(35)	
		MCDBR	29.8	28.1	27.1	24.8	24.2	24.7	25.3	27.3	29.6	29.4	30.1	30.1	(36)	
	5% WB	MCWBR	8.7	8.3	8.7	9.7	10.7	12.5	12.6	13.3	13.3	11.4	9.6	9.3	(37)	
		MCWBR	23.0	22.4	21.6	20.5	20.5	21.5	23.1	24.4	25.8	25.8	25.4	24.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.323	0.313	0.290	0.269	0.255	0.255	0.293	0.319	0.326	0.300	0.302	(40)	
		taud	2.550	2.522	2.529	2.583	2.626	2.636	2.624	2.475	2.374	2.395	2.540	2.574	(41)	
	Ebn at Noon		325	317	310	302	294	292	297	296	302	312	328	331	(42)	
		Edh at Noon	35	35	32	27	23	22	23	30	37	39	35	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2530	2297	1969	1530	1221	1072	1178	1467	1877	2257	2558	2698	(44)	
	RadStd		129	128	87	69	51	60	47	76	93	69	109	125	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page